

CLEAN AIR, DEVELOPING MINDS

From Atmospheric Observation to Human Consequence

TA-14 Authority Governance Institution | Independent scientific evidence reference | Version 1.0 | 8 October 2026

STATUS: PRE-EXAMINATION / BASELINE NOT FROZEN / NO INSTITUTIONAL ENDORSEMENT

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01 | Why this research matters

Columbia University Mailman School of Public Health reported that declining prenatal exposure to air pollution in New York City coincided with improved measures of early childhood cognitive development across successive cohorts. The findings invite a consequential governance question: how should evidence travel from atmospheric observation to scientific interpretation, and then to public decisions without overstating what it establishes?

02 | What the published university summary reports

Evidence element	Reported finding
Study population	More than 1,000 mother-child pairs in long-term studies, 1998-2020
PM2.5	32% decline in annual average residential prenatal concentrations, 1998-2016
NO2	30% decline in annual average residential prenatal concentrations, 1998-2016
PAHs	60% decline in prenatal personal-monitoring concentrations, 1998-2020
Cognitive measure	26% higher age-three Mental Development Index scores, earliest vs latest cohort

The university reports adjustment for maternal education, child sex, gestational age, and household smoking. The summary describes associations across cohorts; it does not establish that the policy changes alone caused the measured cognitive differences. Percentages refer to different measures and observation periods and must not be combined as if they share one denominator.

03 | Evidence boundaries and uncertainty

The research involves different exposure measures, study periods, and sequential cohorts. Interpretation must preserve exposure-estimation methods, monitoring provenance, cohort definitions, assessment instruments, adjustment choices, uncertainty, and plausible confounding. An observed cohort difference is not a randomized policy-effect estimate. A public-health claim must remain within the limits of the underlying study.

04 | The TA-14 governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

For this educational scenario, a proposed consequence might be a policy recommendation, public communication, investment decision, or intervention. The scientific publication is evidence to examine, not an automatic authorization for any particular decision.

05 | Evidence-to-consequence chain

REALITY - Ambient pollution, individual exposure, lived conditions, and developmental outcomes.

RECORD - Measurement records, exposure estimates, cohort data, and reported results.

CONTINUITY - Traceability of instruments, methods, samples, time windows, and transformations.

ADMISSIBILITY - Fitness of the evidence for the precise claim and proposed decision.

BINDING - Link the evidence to the decision, responsible actor, applicable authority, and standing.

COMMIT - Determine whether the exact proposed action may proceed under current conditions.

EXECUTION - Implement only the action actually permitted.

OUTCOME - Observe results, preserve uncertainties, and update the next evidence record.

06 | Interactive examination design for the public showroom

Visitors should examine three clearly distinct propositions: (A) air pollution exposure decreased in the observed periods; (B) cognitive scores differed across cohorts in association with exposure declines; and (C) a specific policy action is presently authorized. Each proposition requires its own evidence and claim ceiling. The interactive choices are ALLOW / HOLD / DENY / ESCALATE; these are teaching dispositions, not findings about Columbia or the researchers.

07 | Eight numbered teaching visuals and narration plan

01 | The human question. Why early-life exposure and cognitive development matter.

- 02 | What was measured. PM2.5, NO2, PAHs and differing observation methods.
- 03 | Time and cohort. Successive cohorts, measurement periods, and comparability.
- 04 | What the results say. Reported reductions and cognitive-score association.
- 05 | What they do not say. Causality, confounding, and claim limitations.
- 06 | The evidence chain. Reality through admissibility, with provenance preserved.
- 07 | The authority boundary. Who can act, under which authority, with what standing NOW.
- 08 | From decision to outcome. ALLOW/HOLD/DENY/ESCALATE, implementation, and feedback.

Each showroom visual should carry its own number and concise instructional labels. Samantha narration should reference the visual by number, with independent scrolling and no automatic movement. The visuals must not depict fabricated data, fictitious study participants as actual subjects, or institutional endorsement.

08 | Proposed next steps, not completed findings

Verify the full peer-reviewed paper and supplemental methods; preserve the original public sources; define a bounded examination question; register the governance if an external party requests a formal TA-14 examination; freeze the baseline; record evidence and uncertainty; and only then issue a disposition supported by the examination record. Until those steps occur, status remains PRE-EXAMINATION / HOLD.

Sources and attribution

Columbia University Mailman School of Public Health, "Clean Air Policies Linked to Improved Cognition Among New York City Children," 17 August 2026.

<https://www.publichealth.columbia.edu/news/clean-air-policies-linked-improved-cognition-among-new-york-city-children>

Research article, Environmental Health (linked by Columbia):

<https://link.springer.com/article/10.1186/s12940-026-01328-1>

TA-14 Authority Governance Institution: <https://www.ta14exchange.com/>

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